

Hexagonal Constructor – Documentation

v2.3

1. Overview

Hexagonal Constructor is a procedural hex-grid generator for Unity featuring:

- Randomized generation (BranchWalk, ClusterGrowth)
 - Shape generation (Disk, Ring, Spiral, Triangle, Rhombus, Rectangle)
 - Custom editor with preview and debugging tools
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2. Installation + Quick Start

1. Import the package through **Package Manager**.
 2. Add component **GridGenerator.cs** to any GameObject.
 3. Click "**Add Required Components**" if prompted.
 4. Assign a hex prefab in **Grid Settings**.
 5. Choose a **generation mode** (Shapes / Randomized).
 6. Choose a **generator algorithm** on the selected generation mode.
 7. **Press Play** to generate.
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3. Inspector Summary

3. 1. Grid Settings

3.1. 1. *Tile Visuals*

- **Hex Prefab** – prefab with SpriteRenderer
- **Hex Scale** – visual scale

3. 1. 2. *Tile Geometry*

- **Hex Orientation** – FlatTop / PointyTop
- **Hex Radius** – hex tile radius

3. 2. Generation Settings

- **Start Axial** – starting hex tile coordinates (Q, R)
- **Generation Mode** – Randomized / Shapes
- **Generator** – depends on the selected mode:
Shape Generator: Disk, Ring, Spiral, Triangle, Rhombus, Rectangle
Randomized Generator: BranchWalkGenerator, ClusterGrowthGenerator

3. 2. 1. *Randomized Generation*

- **Random Algorithm** – BranchWalk / ClusterGrowth
- **Hex Count** – tile count
- **Seed options** – deterministic generation

3. 2. 2. *Shape Generation*

- **Disk / Ring** – radius
 - **Spiral** – spiral length, start direction, growth
 - **Triangle** – side length
 - **Rhombus** – side length, start direction
 - **Rectangle** – width, height, isPointy checkbox
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4. Editor Tools (Optional)

4. 1. Preview Settings

- **Is Active** – toggles preview
- **Hex Color** – gizmo hexes color
- **Hex Scale** – gizmo hex scale

4. 2. Debug Settings

- **Is Active** – toggles generator debug
 - **Step Delay** – time between step-by-step hex generation
 - **Rebuild/Clear Generation** – generates tiles in Scene View
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5. Notes

- Sprite rotation is manual for matching orientation
 - Grid logic is fully axial
 - Tested in Unity 2021.3+
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6. Troubleshooting

Tiles overlap → adjust radius / scale

Orientation looks wrong → rotate prefab (90°)